

MCQs Set 'A'

1. (d) 11250
2. (d) 120°
3. (b) $FR\theta$
4. (c) remains unaffected
5. (a) v
6. (d) $E_n \propto \frac{1}{L_n^2}$
7. (c) amplitude
8. (d) more intense first and less intense later
9. (a) go on increasing
10. (b) IBL along the negative z-direction
11. (c) At first magnitude of velocity decreases and then increases

MCQs Set 'B'

Group 'A'

Set 'B'

1. (a) 11250
2. (a) 141°
3. (c) $\tau\theta$
4. (b) Decrease by a factor of $\sqrt{2}$
5. (a) v
6. (b) $T_n \propto \frac{1}{n}$
7. (a) amplitude
8. (d) more intense first and less intense later
9. (b) 60 cm
10. (d) IBL along the negative z-direction
11. (b) magnitude of velocity goes on increasing

Group 'B'

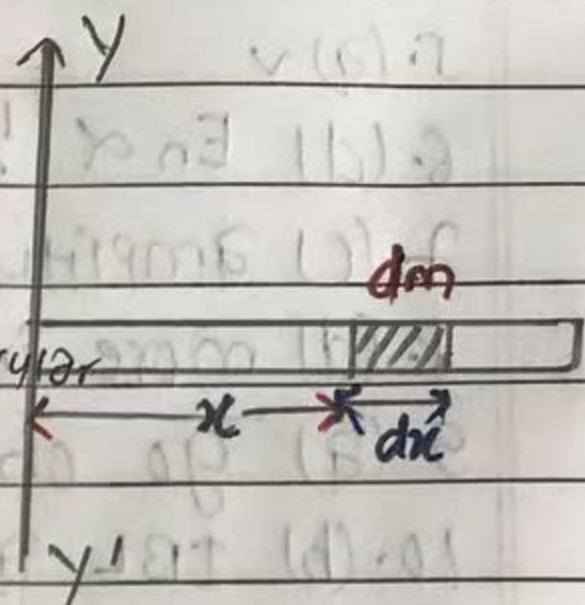
Q. N. 12. (a) Ans: No, the moment of inertia depends only on the mass distribution relative to the axis of rotation & is independent of the state of motion.

12.(b) Ans: NO, the radius of gyration of a body isn't a constant quantity. It depends on:

- Axis of rotation
- Distribution of mass

12.(c) Solution:

Consider a uniform rod of mass 'm' and length 'l' rotating about an axis YY' passing through one end and perpendicular to the rod.



Now,

moment of inertia of dm ,

$$dI = dm x^2 = \frac{m}{l} x^2 dx$$

from 0 to l , Integrate

$$I = \frac{m}{l} \int_0^l x^2 dx$$

$$= \frac{m}{l} \left[\frac{x^3}{3} \right]_0^l$$

$$= \frac{m}{l} \frac{l^3}{3}$$

$$\therefore I = \frac{Ml^2}{3}$$

Also, If the axis passes through the center,

$$\therefore I = \frac{Ml^2}{12}$$

13.(a) Ans: Principle of conservation of angular momentum states that, "If no external torque acts on a system, total angular momentum of system

remains conserved."

Mathematically, $\tau = 0$

$$\text{or, } \frac{dL}{dt} = 0$$

$$\text{Integrating, } \int \frac{dL}{dt} = \int 0$$

$$L = \text{constant}$$

$$\Rightarrow I\omega = \text{constant}$$

$$\therefore I_1\omega_1 = I_2\omega_2$$

for eg: a boiled egg and a raw egg by spinning them on a smooth table.

13.(b) Ans: The middle seat is near the centre of mass of the bus where vertical motion due to bumps is minimised resulting in smooth ride.

13.(c) solution:

$$\text{Moment of inertia (I)} = 0.32 \text{ kg m}^2$$

$$\text{Angular velocity } (\omega) = 120 \text{ rad sec}^{-1}$$

$$\text{Power (P)} = 50 \text{ W}$$

We have,

$$(i) \text{ kinetic energy (K.E.)} = \frac{1}{2} I\omega^2 = \frac{1}{2} \times 0.32 \times (120)^2$$

$$= 2304 \text{ J} //$$

Also,

$$\text{Angular momentum (L)} = I\omega = 0.32 \times 120 = 38.4 \text{ kg m}^2 \text{ s}^{-1} //$$

(ii) frictional couple opposing rotation, $P = \tau\omega$

$$\Rightarrow \tau = \frac{P}{\omega} = \frac{50}{120} = 0.4167 \text{ Nm} //$$

Q.N. 14. (a) Ans: The characteristics of a simple harmonic motion are:

- (i) It is a periodic motion.
- (ii) Acceleration is directly proportional to displacement from mean position. $\Rightarrow a \propto -y$

14. (b) solution:

Let us consider a body of mass 'm' moves in simple harmonic motion. For SHM, acceleration (a) = $-\omega^2 y$

$$\text{Small work done } (dw) = -F dy = -(-ky) dy = ky dy$$

$$\therefore (dw) = \therefore dw = m\omega^2 y dy$$

From 0 to y, Integrate

$$P.E. = \int_0^y dw = m\omega^2 \int_0^y y dy = \frac{1}{2} m\omega^2 y^2$$

$$K.E. = \frac{1}{2} mv^2 = \frac{1}{2} m (\omega \sqrt{A^2 - y^2})^2 = \frac{1}{2} m\omega^2 (A^2 - y^2)$$

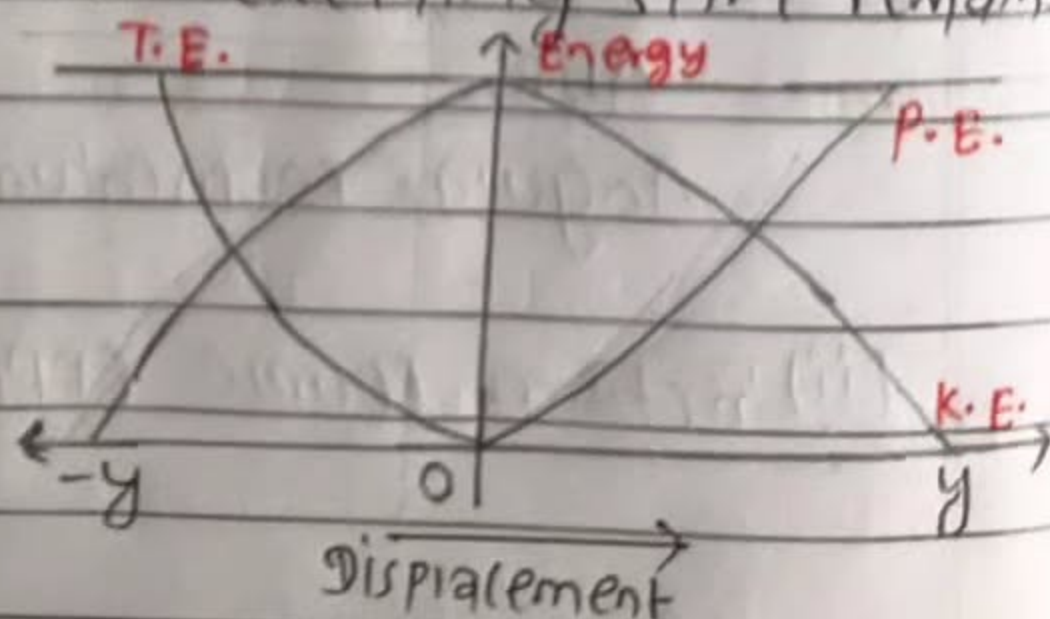
$$\text{Now, Total Energy (T.E.)} = P.E. + K.E.$$

$$= \frac{1}{2} m\omega^2 y^2 + \frac{1}{2} m\omega^2 (A^2 - y^2)$$

$$= \frac{1}{2} m\omega^2 A^2$$

$$\therefore T.E. = \frac{1}{2} kA^2 \quad (\because k, A \text{ are constants})$$

Hence, total energy of a particle executing SHM remains constant.



14. (C) Solutions

Amplitude (A) = 0.1 m

Displacement (x) = 0.05 m

mass (m) = 0.2 kg

Speed (v) = 0.6 m/s

We have,

$$\text{Angular velocity } (\omega) \Rightarrow \omega^2 = \frac{v^2}{A^2 - x^2} = \frac{(0.6)^2}{(0.1)^2 - (0.05)^2} = 48 \text{ rad}^2/\text{sec}$$

Also,

$$\text{Total energy} = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} \times 0.2 \times 48 \times (0.1)^2 = 0.048 \text{ J}$$

Q.N.15. (a) Ans: The principle of superposition of waves states that, "the resultant displacement of the particle is equal to the vector sum of individual displacements due to different waves." $\Rightarrow y = y_1 + y_2 + y_3 + \dots + y_n$

Also,

When two waves of equal amplitude and same frequency travelling in opposite direction superimpose to each other, they give rise to a stationary wave.

Along positive x-axis,

$$y_1 = a \sin(\omega t - kx) \dots (1)$$

Along negative x-axis, $y_2 = a \sin(\omega t + kx) \dots (2)$

Now,

$$\text{resultant wave } (y) = y_1 + y_2 = a \sin(\omega t - kx) + a \sin(\omega t + kx)$$

$$\text{or, } y = a [\sin(\omega t - kx) + \sin(\omega t + kx)]$$

$$\text{or, } y = 2a \cos kx \cdot \sin \omega t$$

$$\therefore y = A \sin \omega t$$

This is the eqn. of stationary wave of amplitude $A = 2a \cos kx$ which varies with x .

15. (b) Solution: Wavelength (λ) = 20 cm

(i) from graph,

$$V_{\max} = 5 \text{ cm/s}$$

$$T = 4 \text{ sec}$$

we have,

$$\text{Angular frequency } (\omega) = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2} \text{ rad/sec}$$

Now,

$$V_{\max} = A\omega$$

$$\therefore A = \frac{V_{\max}}{\omega} = \frac{5}{\pi/2} = \frac{10}{\pi} \text{ cm} //$$

(ii) $\lambda = 20 \text{ cm}$,

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{20} = \frac{\pi}{10} \text{ rad/cm}$$

we have,

$$y = -A \cos(\omega t - kx)$$

$$\text{or, } y = -\frac{10}{\pi} \cos\left(\frac{\pi}{2}t - \frac{\pi}{10}x\right)$$

$$\text{or, } y = \frac{10}{\pi} \cos\left(\frac{\pi}{10}x - \frac{\pi}{2}t\right)$$

$$\therefore y = \frac{10}{\pi} \sin\left(\frac{\pi}{2}t - \frac{\pi}{10}x - \frac{\pi}{2}\right) //$$

Q. N. 16: (a) Ans: The expression for the magnitude of the magnetic Lorentz force is, $F = qvB \sin\theta$

Where, $F \rightarrow$ Magnetic Lorentz force

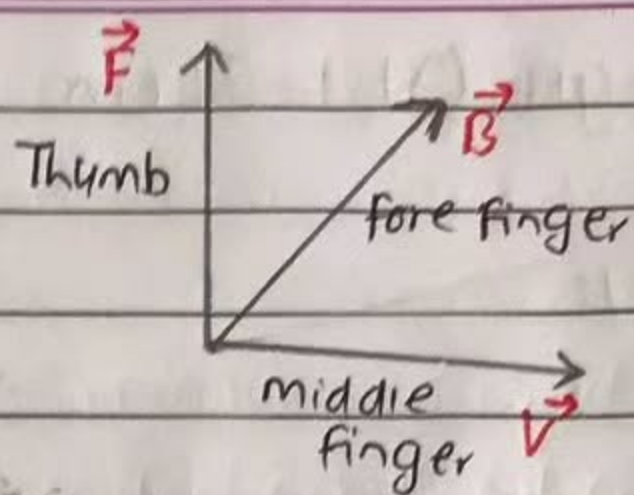
$q \rightarrow$ Charge of particle

$v \rightarrow$ velocity

$B \rightarrow$ Magnetic field

$\theta \rightarrow$ angle between v and B

P.T.O. $\rightarrow$



According to Fleming's left hand rule, for mutually perpendicular, Fore finger points to direction of magnetic field, middle finger points in the direction of motion of positively charged particle ($\vec{v}$), then thumb points to the direction of force.

16. (b) solution:

Magnetic field (B) = 0.500 T

(i) Acceleration, $t=0$

$$\vec{F} = q (\vec{v} \times \vec{B}) = q \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1.95 \times 10^5 & 0 & 2 \times 10^5 \\ 0.500 & 0 & 0 \end{vmatrix}$$

$$\therefore \vec{F} = 1.6 \times 10^{-19} \times 1.6 \times 10^5 = 1.6 \times 10^{-14} \text{ N}$$

$$\Rightarrow \therefore \vec{a} = \frac{\vec{F}}{m_p} = \frac{1.6 \times 10^{-14}}{1.67 \times 10^{-27}} = 9.58 \times 10^{12} \hat{j} \text{ m/s}^2 //$$

(ii) yes, the path is helical.

$$\text{Radius of helix (r)} = \frac{mv_y}{qB} = \frac{1.67 \times 10^{-27} \times 2.00 \times 10^5}{1.6 \times 10^{-19} \times 0.500}$$

$$\therefore r = 4.18 \times 10^{-3} \text{ m} //$$

Also, $v_y = v_x = 1.95 \times 10^5 \text{ m/s}$, $v_z = v_z = 2.00 \times 10^5 \text{ m/s}$

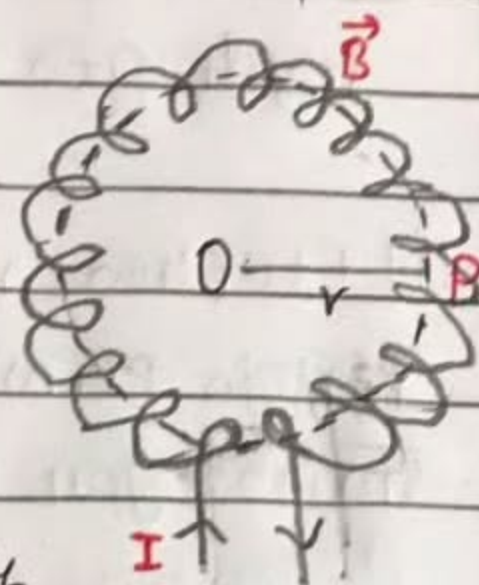
$$\text{pitch of helix (P)} = v_x T = v_y \frac{2\pi R}{v_z} = \frac{1.95 \times 10^5 \times 2\pi \times 4.18 \times 10^{-3}}{2 \times 10^5}$$

$$\therefore P = 2.56 \times 10^{-2} \text{ m} //$$

P.T.O. $\rightarrow$

Q.N. 17. (a) Ans: An endless solenoid in the form of a ring is called toroid.

Consider an air-cored toroid with radius 'r' having 'n' turns per unit length and carrying current 'I'.



The line integral $\oint \vec{B} \cdot d\vec{l}$ over closed path,

$$\oint \vec{B} \cdot d\vec{l} = B dl \cos 0^\circ = B \times 2\pi r \quad (\because \text{angle between } d\vec{l} \text{ and } \vec{B} = 0)$$

According to Ampere's law,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \times \text{total current}$$

Let toroid has N turns. Then, $\oint \vec{B} \cdot d\vec{l} = \mu_0 (NI)$

$$\text{or, } B 2\pi r = \mu_0 NI$$

$$\text{or, } B = \mu_0 \left(\frac{N}{2\pi r} \right) I$$

Where, $n = \frac{N}{2\pi r}$ = no. of turns per unit length

$$\therefore B = \mu_0 n I$$

17. (b) Solution:

$$I_1 = 6 \text{ A (into the page)}$$

$$I_2 = -4 \text{ A (out of the page)}$$

$$\text{value of integral } \oint \vec{B} \cdot d\vec{l} = \mu_0 I = 3.77 \times 10^{-6} \text{ Tm}$$

Now,

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_1 + I_2 + I_3)$$

$$\text{or, } 3.77 \times 10^{-6} = 4\pi \times 10^{-7} (6 - 4 + I_3)$$

$$\text{or, } \frac{3.77 \times 10^{-6}}{4\pi \times 10^{-7}} = 2 + I_3$$

$$\text{or, } 3 = 2 + I_3$$

$$\therefore I_3 = 1 \text{ A} //$$

Q.N.18. (a) Ans: The postulates of Bohr's theory of hydrogen atom:

(i) Electron revolves around the nucleus only in a certain permitted orbit where its angular momentum is an integral multiple of $\frac{h}{2\pi}$.

$$\text{i.e. } mvr = \frac{nh}{2\pi} \quad \text{where } n = 1, 2, 3, \dots$$

(ii) When an electron jumps higher to lower orbit or vice-versa, energy is emitted in form of electromagnetic wave. $\therefore E_2 - E_1 = hf$

(iii) The electrostatic force between nucleus and electron provides necessary centripetal force.

$$\therefore \frac{mv^2}{r} = \frac{Ze^2}{4\pi\epsilon_0 r^2}$$

18. (b) Solution:

Speed of electron in n^{th} orbit of atom,

$$v_n = \frac{Ze^2}{2\epsilon_0 nh}$$

for hydrogen, $Z = 1$

$$\therefore v_n = \frac{e^2}{2\epsilon_0 nh} \quad \dots (1)$$

we have,

$$\alpha = \frac{e^2}{2\epsilon_0 hc}$$

$$\Rightarrow \frac{e^2}{2\epsilon_0 h} = \alpha c$$

Substitute this value in eqn (1),

$$\text{Or, } v_n = \frac{\alpha c}{n} \quad \left(\because \alpha \approx \frac{1}{137} \right)$$

$$\text{Or, } v_n = \frac{1}{137} \cdot \frac{c}{n}$$

For 1st excited state, $n = 2$

$$\therefore v_n = \frac{1}{137} \cdot \frac{c}{2} = \frac{1}{274} c \quad (\text{where } c = \text{speed of light}) //$$

18.(c) Ans: we have,

$$v_n = \frac{e^2}{2\epsilon_0 n h}$$

$$\Rightarrow v_n \propto \frac{1}{n}$$

In outer orbit, n is larger. So, v_n is smaller. But, in inner orbit, n is smaller. So, v_n is larger.

Q.N.19. (a) solution:

$$\text{Radius } (r) = 7 \times 10^{-7} \text{ m}$$

$$\text{Density } (\rho) = 900 \text{ kg m}^{-3}$$

$$\text{Potential difference } (V) = 500 \text{ V}$$

$$\text{we have, } d = 6 \text{ mm} = 6 \times 10^{-3} \text{ m}$$

$$qE = mg$$

$$\text{Or, } q = \frac{mg}{E} = \frac{\frac{4}{3} \pi r^3 \rho g}{V/d} = \frac{4 \pi r^3 \rho g d}{3V}$$

$$\text{Or, } q = \frac{4 \pi \times (7 \times 10^{-7})^3 \times 900 \times 10 \times 6 \times 10^{-3}}{3 \times 500}$$

$$\therefore q = 1.55 \times 10^{-19} \text{ C} //$$

19.(b) Ans: Discrepancy is due to the error in the measurement of radius of oil drop which is in the order of 10^{-7} .

19. (c) Ans: If the potential difference is raised above 500V, oil drop accelerates upward.

19. (d) Ans: If the polarity of the plates is reversed while keeping potential difference at 500V, oil drop accelerates downward causing electric force to act downwards in same direction as gravity.

Group 'C'

Q.N.20. (a) Ans: According to Laplace, heat cannot be exchanged with surroundings during compression and rarefaction because propagation of sound in air is very fast process and air is bad conductor of heat. Thus, it is an adiabatic process.

$$\text{So, } pV^\gamma = \text{constant}$$

On differentiating,

$$p\gamma V^{\gamma-1} dV + V^\gamma dp = 0$$

$$\text{or, } p\gamma dV + V dp = 0$$

$$\text{or, } \gamma p = - \frac{dp}{(dV/V)}$$

$$\therefore \gamma p = \beta$$

$$\text{Also, velocity of sound in gas is, } v = \sqrt{\frac{\beta}{\rho}}$$

$$\therefore v = \sqrt{\frac{\gamma p}{\rho}} \dots (1)$$

$$\text{We have, } pV = nRT$$

$$\text{or, } pV = \frac{m}{M} RT \quad \left(\because n = \frac{m}{M} \right)$$

$$\text{or, } \frac{pV}{m} = \frac{RT}{M} \quad \left(\because \rho = \frac{m}{V} \right)$$

$$\text{Or, } \frac{P}{\rho} = \frac{RT}{M}$$

from eqn (1),

$$\therefore v = \sqrt{\frac{\gamma RT}{M}}$$

Since, γ , M and R are constants for a gas,

$$\therefore v \propto \sqrt{T}$$

20. (b) Solution:

At 0°C ,

$$v_0 = 332 \text{ m/s}$$

$$T_0 = 0^\circ\text{C} = 273 \text{ K}$$

At 25°C ,

$$v_{25} = ?$$

$$T_{25} = 25^\circ\text{C} = 25 + 273 = 298 \text{ K}$$

We have,

$$v_{25} = v_0 \sqrt{\frac{T_{25}}{T_0}}$$

$$\text{Or, } v_{25} = 332 \sqrt{\frac{298}{273}}$$

$$\therefore v_{25} = 346.87 \text{ m/s} //$$

Now,

$$\Delta t = 3 \text{ sec}$$

$$\Delta t = t_1 - t_2$$

$\therefore t_2 \rightarrow \text{speed}$

$$\text{Or, } 3 = \frac{d}{v_{25}} - \frac{d}{c} \quad (\because c \rightarrow \text{speed of light})$$

$$\text{Or, } 3 = \frac{d}{346.87} - \frac{d}{3 \times 10^8}$$

$$\text{Or, } 3 = d \left(\frac{1}{346.87} - \frac{1}{3 \times 10^8} \right)$$

$$\therefore d = 1040.61 \text{ m} //$$

P.T.O. $\rightarrow$

20. (c) Ans: As $v \propto \frac{1}{\sqrt{\rho}}$, increased humidity leads to

decrease in air density causes the speed of sound in air to increase.

20. (d) Ans: The notes whose frequencies are integral multiples of the fundamental frequency are called harmonics.

Also,

Harmonics are always above fundamental frequency except for fundamental itself. Thus, all harmonics are overtones. But, some overtones may not be integral multiples of fundamental frequency. Thus, all overtones are not harmonics.

Q.N. 21. (a) Ans: A current element produces a magnetic field even though it is electrically neutral because it involves movement of electric charges (electrons) and moving charges generate magnetic field.

21. (b) Ans: Biot-Savart law states that, "Magnetic field produced by a current carrying conductor

(i) directly proportional to current ($dB \propto I$)

(ii) directly proportional to length of current element ($dB \propto dl$)

(iii) directly proportional to sine of angle $\vec{dl}$ and $\vec{r}$ ($dB \propto \sin \theta$)

(iv) inversely proportional to square of distance between current element and point

$$(dB \propto \frac{1}{r^2})$$

P.T.O. $\rightarrow$

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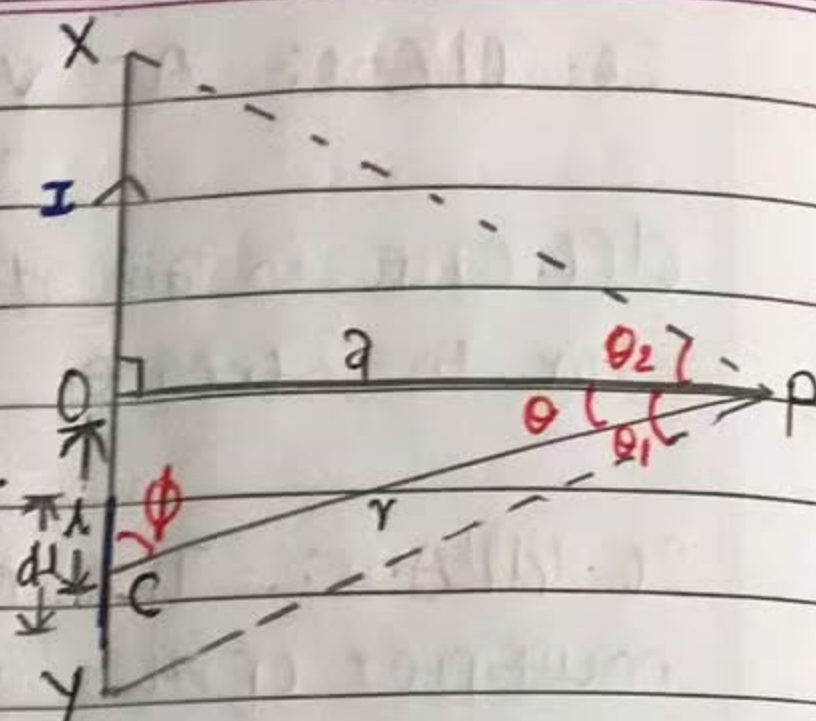
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- (iv) inversely proportional to square of distance between current element and point ($dB \propto \frac{1}{r^2}$)

21. (c) Ans:

Let P be a point at perpendicular distance 'a' from a long straight conductor XY carrying current 'I'.
Let ϕ be angle between dl and r .



According to Biot-Savart law, magnetic field at P due to dl ,

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \phi}{r^2} \quad \dots (1)$$

In ΔOPC , $\sin \phi = \frac{a}{r} = \cos \theta$

$$\therefore r = \frac{a}{\cos \theta} = a \sec \theta \quad \dots (2)$$

Also, $\tan \theta = \frac{l}{a} \Rightarrow l = a \tan \theta$

Differentiating, $dl = a \sec^2 \theta d\theta \quad \dots (3)$

$$\sin \phi = \sin (90^\circ - \theta) = \cos \theta \quad \dots (4)$$

Using eqns (2), (3) and (4) in eqn (1),

$$dB = \frac{\mu_0}{4\pi} \frac{I a \sec^2 \theta d\theta \cos \theta}{(a \sec \theta)^2}$$

$$\therefore dB = \frac{\mu_0 I}{4\pi a} \cos \theta d\theta$$

Integrating,

$$B = \int dB = \int_{\theta_1}^{\theta_2} \frac{\mu_0 I}{4\pi a} \cos \theta d\theta = \frac{\mu_0 I}{4\pi a} \int_{\theta_1}^{\theta_2} \cos \theta d\theta$$

$$\therefore B = \frac{\mu_0 I}{4\pi a} [\sin \theta]_{\theta_1}^{\theta_2}$$

$$\therefore B = \frac{\mu_0 I}{4\pi a} (\sin \theta_2 + \sin \theta_1)$$

for infinitely long conductor, $\theta_1 = \theta_2 = \frac{\pi}{2}$

$$B = \frac{\mu_0 I}{4\pi a} \left(\sin \frac{\pi}{2} + \sin \frac{\pi}{2} \right) = \frac{\mu_0 I}{4\pi a} (1+1)$$

$$\therefore B = \frac{\mu_0 I}{2\pi a}$$

21. (d) Solution:

$$\text{Radius (r)} = 2.40 \text{ cm} = 0.024 \text{ m}$$

$$\text{Number of turns (N)} = 800 \text{ turns}$$

$$\text{Magnetic field (B)} = 0.0520 \text{ T}$$

(i)

we have,

$$B = \frac{\mu_0 N I}{2r}$$

$$\text{or, } I = \frac{2rB}{\mu_0 N} = \frac{2 \times 0.024 \times 0.0520}{4\pi \times 10^{-7} \times 800}$$

$$\therefore I = 2.76 \text{ A} //$$

(ii)

we have,

Magnetic field on the axis = $\frac{1}{8}$ magnetic field at centre

$$\frac{\mu_0 N I r^2}{2(x^2 + r^2)^{3/2}} = \frac{1}{8} \frac{\mu_0 N I}{2r}$$

$$\text{or, } \frac{r^2}{(x^2 + r^2)^{3/2}} = \frac{1}{8r}$$

$$\text{or, } 8r^3 = (x^2 + r^2)^{3/2}$$

$$\text{or, } 8^{2/3} \cdot r^{3 \cdot 2/3} = x^2 + r^2$$

$$\text{or, } 4r^2 - r^2 = x^2$$

$$\text{or, } x = \sqrt{3r^2} = \sqrt{3} (0.024)^2$$

$$\therefore x = 0.04157 \text{ m} //$$
$$= 4.157 \text{ cm}$$

Q.N. 22. (a) Ans: The minimum energy of photon required just to liberate an electron from metal surface with zero velocity is called work function.

Also,

$$\text{work function } (\phi) = 6.4 \times 10^{-19} \text{ J}$$

$$\text{frequency } (f) = 1.2 \times 10^{15} \text{ Hz}$$

we have,

$$\text{Energy } (E) \Rightarrow hf = 6.62 \times 10^{-34} \times 1.2 \times 10^{15}$$

$$\therefore E = 7.94 \times 10^{-19} \text{ J} > \phi$$

Hence, photoelectric effect will take place.

22. (b) Ans:

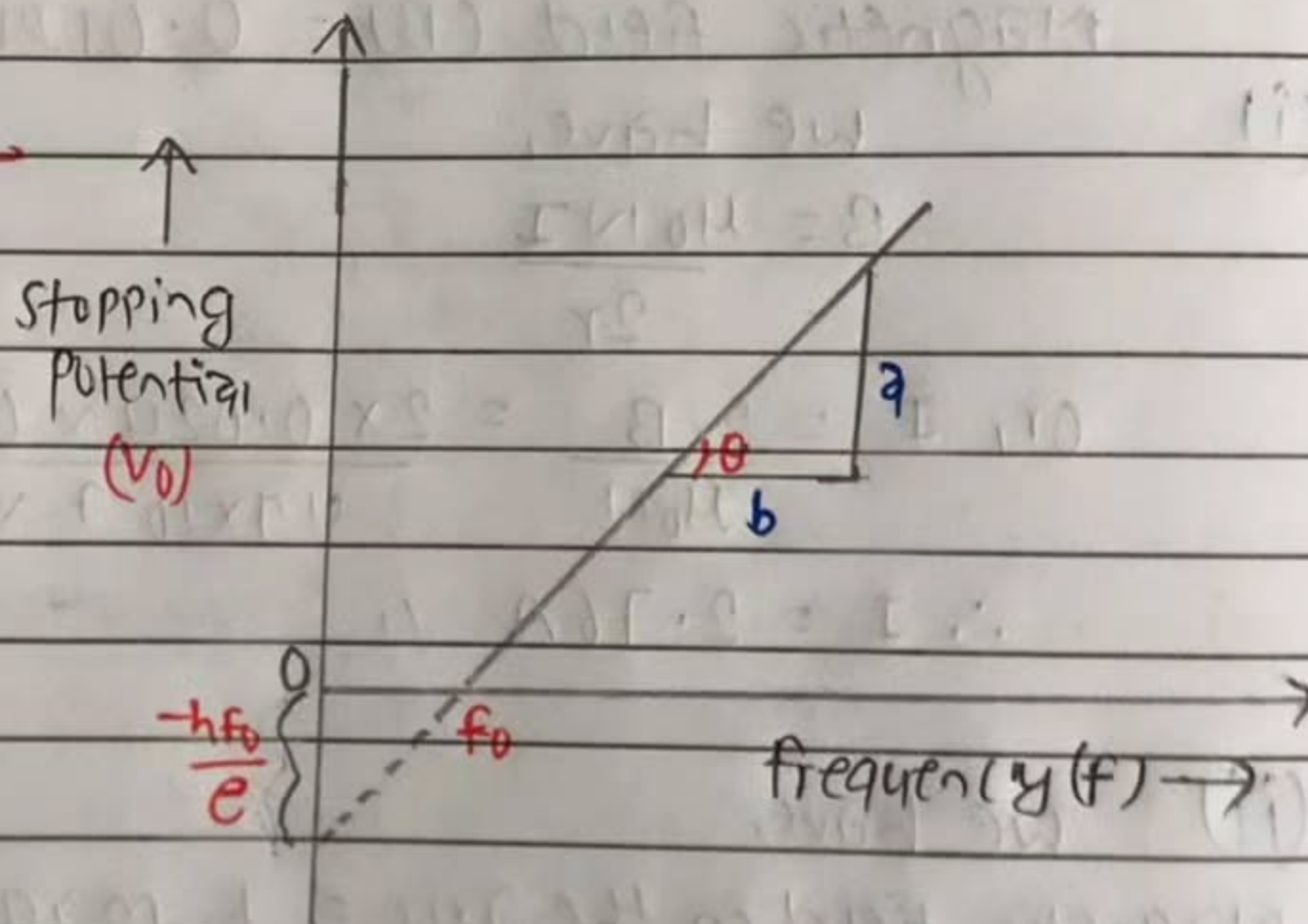


fig: Graph between Stopping Potential and frequency

At stopping potential,

$$eV_0 = \frac{1}{2} mv_{\max}^2 \quad \dots (1)$$

From Einstein's photoelectric equation,

$$\frac{1}{2} mv_{\max}^2 = hf - \phi$$

$$\text{or, } \frac{1}{2} mv_{\max}^2 = hf - hf_0 \quad \dots (2)$$

from eqn (1) & (2),

$$eV_0 = hf - hf_0$$

$$\therefore V_0 = \frac{h}{e} f - \frac{h}{e} f_0$$

Compare with $y = mx + c$,

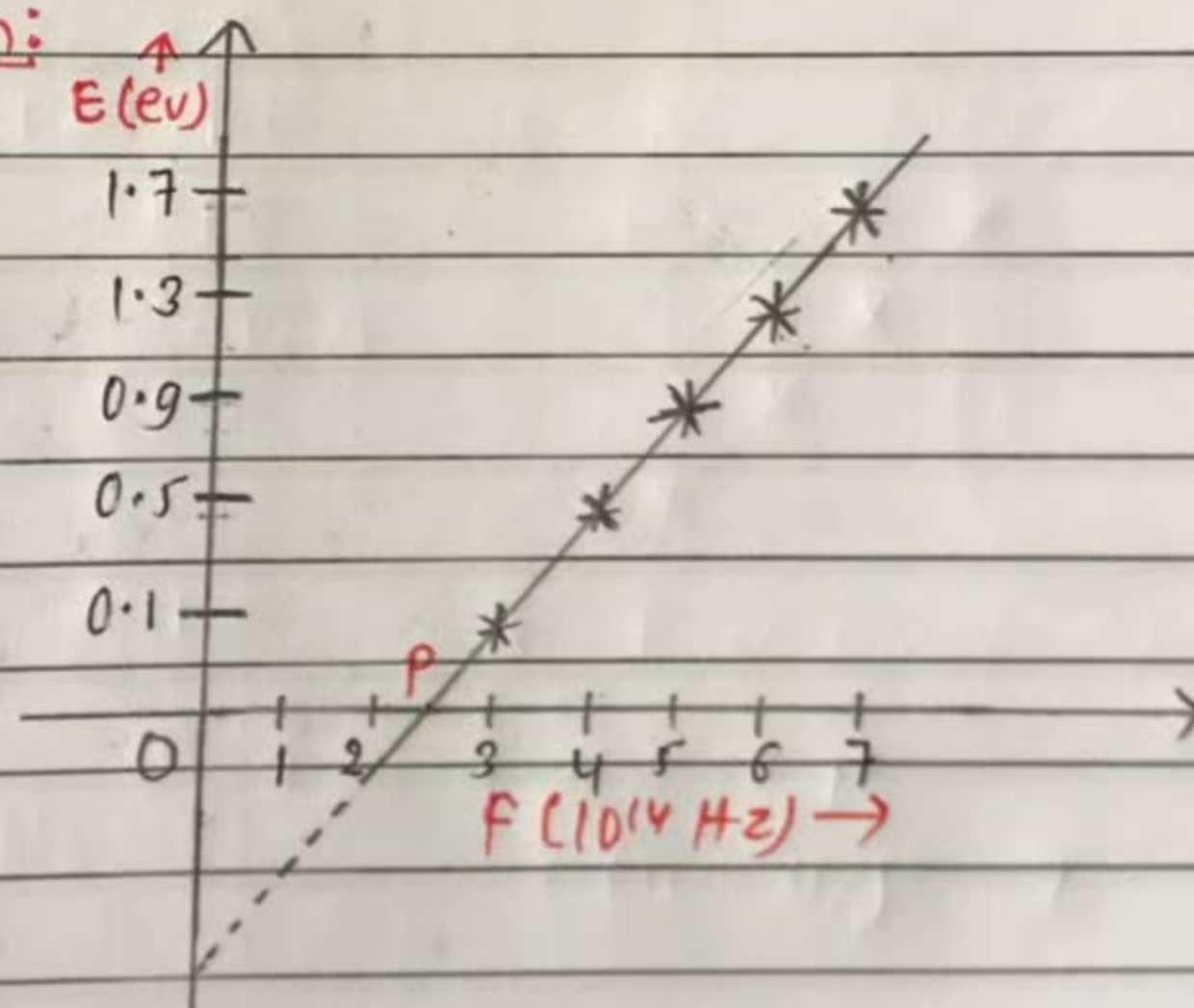
$$\text{Slope } (m) = \frac{h}{e}$$

$$\text{Or, } \tan \theta = \frac{h}{e}$$

$$\therefore h = e \tan \theta$$

In this way, the value of Planck's constant can be determined from the graph.

22.(c) Solution:



From Graph,

$$\begin{aligned} \text{Threshold frequency } (f_0) &= \text{freq. at point P} \\ &= 2.5 \times 10^{14} \text{ Hz} // \end{aligned}$$

$$\begin{aligned} \text{Work function } (\phi) &= hf_0 = 6.62 \times 10^{-34} \times 2.5 \times 10^{14} \\ &= 1.65 \times 10^{-19} \text{ J} // \end{aligned}$$

P.T.O. →

Planck's constant (h) = Slope of graph

$$= \frac{0.5 \times 1.6 \times 10^{-19} - 0.1 \times 1.6 \times 10^{-19}}{4 \times 10^{14} - 3 \times 10^{14}}$$

$$h = 6.4 \times 10^{-34} \text{ Js} //$$

"The END"